

Alan Szmyt

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[LinkedIn](#) | [GitHub](#) | [ORCID](#)

Summary

Software engineer focused on AI-native workflows, platform systems, and scalable software architecture. Experienced designing mission-critical applications, automation frameworks, and knowledge systems spanning backend, mobile, data, and infrastructure domains. Combines systems thinking, software engineering, and human-centered design to transform complex problems into reliable, maintainable solutions.

Software Engineer, Associate Staff

MIT Lincoln Laboratory

Lexington, MA

Apr 2019 – Aug 2025

- Developed full-stack geospatial and decision-support systems for government and research sponsors, integrating data engineering, visualization, and automation technologies to support mission-critical analysis and field operations.
- Led development of a mobile tracking system for USAID to monitor commodities in low-connectivity environments, successfully field-testing pilot shipments to Africa and enabling end-to-end supply chain visibility.
- Delivered a humanitarian deconfliction notification platform enabling analysts to visualize no-strike zones and submit secure, auditable reports, contributing to a follow-on effort that secured approximately \$1M in additional funding.
- Built a counter-human trafficking investigation tool that enabled law enforcement analysts to triage large volumes of digital evidence and reduce time to actionable leads.
- Developed externally accessible geospatial decision-support applications that integrated shared data hubs, interactive mapping, and domain-specific datasets to improve situational awareness and analytical decision-making.
- Contributed to NATO disaster-response exercises by extending field-ready mobile capabilities, including hazard visualization, area-of-interest management, and real-time location reporting across multinational operations.
- Designed reproducible development environments and optimized Android (Gradle, Java/Kotlin) and modern web (Vite/ESBuild) build pipelines, improving onboarding speed, build reproducibility, and developer feedback cycles.

Founder & Systems Architect

Incompris LLC

Wilmington, MA

2023 – Present

- Founded and operate an independent creative-technology studio focused on AI-assisted workflow systems, deterministic publishing pipelines, and multimodal content automation.
- Architected a specification-driven publishing platform that transforms canonical creative assets into reproducible multi-format artifacts (PDF, CBZ, WebP, AVIF, and LaTeX) through staged automation workflows and validation pipelines.
- Designed AI-native workflow architectures that treat language models as bounded system components with explicit contracts, synchronization checkpoints, and reproducible execution paths.
- Built multimodal content-generation pipelines integrating LLMs, image generation, video automation, metadata processing, and publication tooling into cohesive end-to-end workflows.
- Developed local-first development environments and CI-backed automation systems using Docker, GitHub Actions, and reproducible infrastructure patterns to accelerate development and artifact validation.
- Authored reflector, an independent research publication exploring synchronization architectures, recursive AI-assisted software engineering workflows, and human-AI governance systems.

Publications

- **reflector: Reflective Synchronization Systems for Recursive AI-Assisted Software Engineering** (2026) Independent research publication proposing a synchronization architecture for AI-assisted software engineering workflows, introducing concepts including recursive drift, reflective auditing, synchronization checkpoints, and specification-driven orchestration. Published through Zenodo with DOI: [10.5281/zenodo.20477044](https://doi.org/10.5281/zenodo.20477044). [Publication](#) | [ORCID](#)

Education

M.S., Software Development

Boston University

Boston, MA

2023

- Focused on software engineering, full-stack development, and system architecture.

- Capstone projects included a WAV metadata parser in Python, an Android beatmaker application using JNI and C++, and a personalized music recommendation system incorporating algorithmic ranking techniques.

B.S., Computer Science (Minor in Mathematics)

University of Massachusetts Lowell

Lowell, MA

2017

- Academic focus included Operating Systems, Data Structures, Algorithms, Computer Graphics, and Discrete Mathematics.
- Participated in independent research exploring decentralized blockchain networks.

Skills

Languages: Python, TypeScript, Java, Kotlin, Dart, C++, SQL, Bash

Frontend & Mobile: React, Redux Toolkit, RTK Query, Flutter, Android, JavaScript, HTML/CSS

Backend & Data: Flask, PostgreSQL, MySQL, Elasticsearch, GeoServer, REST APIs, GraphQL

Platform & Infrastructure: Docker, Kubernetes, GitHub Actions, Devcontainers, Pulumi, Linux, CI/CD

Geospatial & Visualization: deck.gl, Kepler.gl, GeoJSON, Mapbox, Data Visualization

AI & Automation: LLM Workflows, AI-Assisted Development, Workflow Automation, Specification-Driven Systems, Publication Pipelines